Week 1 Deliverables: Dataset Selection & Annotator Development

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1. Dataset Proposal

Project Title

AI-Based Brain Tumor Detection and Segmentation System Using Deep Learning

Domain

Healthcare and Medical Imaging

Dataset Selected

BRISC 2025 Brain Tumor Dataset

Dataset Source:
Scientific Data Journal

Dataset Download Source:
BRISC 2025 Kaggle Dataset

Overview

The selected dataset focuses on healthcare and medical imaging, specifically Brain MRI (Magnetic Resonance Imaging) scans containing multiple tumor categories. The dataset includes MRI images of:

- Glioma Tumors
- Meningioma Tumors
- Pituitary Tumors
- Healthy Brain Scans (No Tumor)

The dataset is highly suitable for Computer Vision and Deep Learning applications because it supports multiple tasks including image classification, object detection, and semantic segmentation.

The dataset structure already contains organized folders for classification and segmentation tasks, making it ideal for the project requirements.

Project Scope Fulfillment

The selected dataset fulfills all major requirements mentioned in the project scope and timeline.

1. Image Classification

The MRI images will be classified into four categories:

- Glioma
- Meningioma
- Pituitary
- No Tumor

Deep learning models such as ResNet and EfficientNet can be utilized for this task.

2. Object Detection

Tumor regions will be localized using bounding boxes generated through the custom annotation tool.

The annotated dataset will later be used to train object detection models such as:

- YOLOv8
- Faster R-CNN

3. Segmentation

The segmentation component of the dataset contains pixel-level masks that will be utilized in Week 4 for semantic segmentation tasks.

Segmentation models planned for implementation include:

- U-Net
- Attention U-Net

Dataset Cleaning Process

A custom preprocessing and cleaning pipeline was developed using Python and OpenCV to sanitize and standardize the dataset before training.

The cleaning script performs the following operations:

- Recursively scans all dataset directories
- Detects and removes corrupted or unreadable images
- Verifies image integrity using OpenCV
- Standardizes image dimensions
- Resizes all MRI scans to 640×640 resolution
- Organizes cleaned images into structured directories

The standardized image size ensures compatibility with deep learning architectures such as YOLOv8, ResNet, and U-Net.

2. Custom Annotator Development

A custom Medical Image Annotation Tool was developed from scratch using:

- Python
- Tkinter
- OpenCV

The tool was specifically designed for medical image annotation and dataset preparation.

Objectives of the Annotation Tool

The annotator was developed to fulfill the requirement of creating a custom annotation utility for healthcare image labeling.

The tool assists in:

- Tumor localization
- Bounding box generation
- Label creation
- Annotation visualization
- Dataset preparation for object detection tasks

Key Features of the Annotator

Dynamic Dataset Loading

The tool allows users to dynamically select dataset directories using a graphical interface.

Bounding Box Annotation

Users can manually draw tumor bounding boxes using mouse interactions.

Multi-Class Support

The annotator supports multiple tumor classes including:

- Glioma
- Meningioma
- Pituitary
- No Tumor

Persistent Annotation Loading

Previously saved annotations are automatically loaded when revisiting an image.

YOLO Format Export

Annotations are exported in the standard YOLO object detection format.

Visualization Export

The tool additionally exports visualized annotated images for quality validation and reporting purposes.

Segmentation Support

Polygon-based annotation support was added to facilitate future segmentation tasks.

Technologies Used

Technology	Purpose
Python	Core Programming
OpenCV	Image Processing
Tkinter	Graphical User Interface
PIL (Pillow)	Image Rendering
NumPy	Numerical Operations

Annotation Tool Workflow

The workflow of the custom annotation tool is as follows:

```
Load MRI Image
    ↓
Select Tumor Class
    ↓
Draw Bounding Box
    ↓
Save Annotation
    ↓
Export YOLO Labels
    ↓
Generate Visualization Output
```

Annotator Source Code

The annotator source file was implemented as:

annotator.py

The implementation utilizes:

- cv2.rectangle() for bounding box generation
- cv2.polylines() for segmentation polygon drawing
- Dynamic saving to annotation directories

3. 20 Sample Annotations

As part of the Week 1 deliverables, 20 MRI images were manually annotated using the custom annotation tool.

The annotations were generated for object detection dataset preparation.

Annotation Format

The labels were exported using the standard YOLO annotation format:

[class_id] [x_center] [y_center] [width] [height]

All coordinate values are normalized between:

0.0 → 1.0

relative to the image dimensions.

Example Annotation Output

File:

brisc2025_train_00003_gl_ax_t1.txt

Content:

0 0.317969 0.488281 0.132812 0.164062

Explanation of Annotation

Value	Meaning
0	Class ID for Glioma
0.317969	X-coordinate center

Value	Meaning
0.488281	Y-coordinate center
0.132812	Bounding box width
0.164062	Bounding box height

Annotation Validation

To ensure annotation quality:

- Visualized annotated images were exported
- Bounding boxes were manually verified
- Annotation consistency was checked across classes

The visualized samples were stored in:

annotations/images/

Folder Structure

```
BrainTumorProject/
│
├── dataset/
│   ├── classification_task/
│   └── segmentation_task/
│
├── annotations/
│   ├── labels/
│   └── images/
│
├── annotator/
│   └── annotator.py
│
├── outputs/
│
└── clean_dataset.py
```

Conclusion

The Week 1 objectives were successfully completed, including:

- Selection of a healthcare and medical imaging dataset
- Dataset cleaning and preprocessing
- Development of a custom annotation tool
- Creation of 20 manually annotated MRI samples